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Uncertainty Analysis of Static and Dynamic Models of the Upper Permian Utilizing the Big Loop to Enhance Efficiency and Minimize Computational Time

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Abstract

In the rapidly advancing realm of artificial intelligence, machine learning, workflow automation, and cutting-edge hardware technologies, the adoption and integration of these innovations within the oil and gas industry has witnessed substantial growth. This is particularly evident in the domain of subsurface reservoir modelling, where these technologies are being leveraged to enhance the accuracy and efficiency of reservoir simulation and optimization processes.

Subsurface reservoir models are a critical component in field development planning, as they provide essential insights into both the overall field recovery and the individual reservoir performance under a variety of operational scenarios. These models, which are typically constructed as three-dimensional, physics-based representations of the reservoir, serve as powerful tools for conducting "what-if" analyses, helping to maximize the net present value (NPV) of assets and ensuring the highest possible return on investment. The field of subsurface reservoir science encompasses a broad range of specialized domains, including seismic interpretation, seismic inversion, sedimentological analysis, petrophysical interpretation, geological modelling, dynamic modelling, production facilities modelling, feasibility studies, and economic evaluations.

Reservoir models are typically designed as tailored, fit-for-purpose models, integrating data and insights from multiple domains depending on the specific needs of the project. However, as is common with any complex, multi-disciplinary process, the development of reservoir models is often hindered by several challenges related to workflow automation, data consistency across domains, and the time-consuming nature of model building. Some of the key challenges include:

- **Fragmentation:** The absence of seamless integration between data, results, and interpretations across different subsurface modelling domains, leading to fragmented and inefficient workflows.
- **Inconsistency:** In many cases, results and interpretations derived from one domain may be modified or adjusted in subsequent domains for modelling or historical calibration purposes, often without appropriate feedback to the preceding domains. This lack of consistent updates results in

models that are inconsistent across domains, diminishing the overall predictability and reliability of the final model.

- **Deterministic Approach:** Traditionally, reservoir modelling processes conclude with the generation of only a few model realizations, often limited to high, medium, and low-case scenarios, which may not fully capture the range of uncertainties inherent in the subsurface.
- **Time-Consuming:** The lack of automation in the modelling process significantly prolongs the timeline required to build accurate and reliable models, often delaying decision-making processes.
- **Model Updates:** The challenges related to integration and consistency across subsurface domains also complicate the process of updating reservoir models, making it time-consuming task.

This paper introduces an innovative solution in the form of an integrated forward modelling workflow—referred to as the Big Loop—which facilitates the seamless propagation of reservoir uncertainties across multiple subsurface domains. The aim of this workflow is to create more reliable, adaptive, and ensemble-based reservoir models, ensuring a more robust and efficient approach to subsurface reservoir management.

Introduction

The integrated forward modelling approach has been successfully applied to a deep gas field located offshore in Abu Dhabi. This field comprises five stacked carbonate reservoirs, with the top four reservoirs exhibiting hydrodynamic pressure communication, indicating a continuous fluid system between them. In contrast, the lower, fifth reservoir is fully isolated, with a distinct and deeper gas-water contact, making it a separate reservoir unit. The geological model of the field indicates that the reservoir zones are conformable, meaning there are no significant truncations or pinching of the strata. The field itself is characterized by a 4-way dip closure, which provides a natural trapping mechanism for hydrocarbons. The uppermost layer is capped by a dense anhydrite, which acts as an effective cap rock contributing to the integrity of the reservoir system.

The implemented workflow draws inspiration from previous successful implementations, such as those carried out at ADNOC Offshore (Alqallabi S. et al.) and has been implemented for synthetic fractured carbonate case (Abdallah M. et al.), which demonstrated seamless integration and efficient uncertainty propagation across various subsurface reservoir modelling domains. This workflow spans from seismic interpretation through geological modelling to dynamic simulation, ensuring a comprehensive and cohesive approach to subsurface modelling. This workflow has proven highly effective in addressing several of the challenges previously identified, demonstrating robust capabilities in the following areas:

- **Consistency and Seamless Integration:** One of the primary strengths of this integrated workflow is its ability to maintain consistency across all subsurface modelling domains. By ensuring that each domain—whether seismic, geological, or dynamic—is fully aligned, the final model used for field development planning remains consistent throughout the entire process. This eliminates the need for arbitrary adjustments or unjustified geological modifications during the history calibration phase, ensuring that the model reflects a true and accurate representation of the subsurface without introducing inconsistencies. The seamless integration between domains significantly reduces the potential for errors, ensuring a high level of reliability in the final model.
- **Ensemble-Based Approach:** The workflow employs an ensemble-based methodology, where an integrated uncertainty propagation process generates a calibrated ensemble of models. This allows for a more accurate estimation of recovery probabilities, considering the inherent geological uncertainties that may impact the reservoir's performance. By incorporating this ensemble approach, the workflow provides a more comprehensive and probabilistic understanding of reservoir behaviour, enabling better-informed decision-making and more effective optimization of field development strategies under various geological uncertainties.

- **Evergreen Workflow:** A standout feature of the implemented workflow is its ability to remain adaptable and evergreen over time. As new data becomes available—whether from drilling new wells, recording additional petrophysical logs, or obtaining updated production data—the workflow allows for the seamless incorporation of this information into each relevant modelling domain. This collaborative connected approach ensures that all domains remain synchronized, and the entire reservoir model can be easily updated with minimal time and effort.

Methodology

The methodology begins with the development of a repeatable and integrated geological modelling workflow (Figure 1), designed to automate the construction of comprehensive geological models using all available subsurface data and values of model uncertainty variables. Each subsurface reservoir modelling domain—such as seismic interpretation, geological modelling, and petrophysical analysis—is systematically reviewed, and the associated data is subjected to quality control and validation procedures. This allows for the quantification of inherent uncertainties and the establishment of a multi-dimensional, multi-domain uncertainty space that forms the foundation for subsequent model realizations.

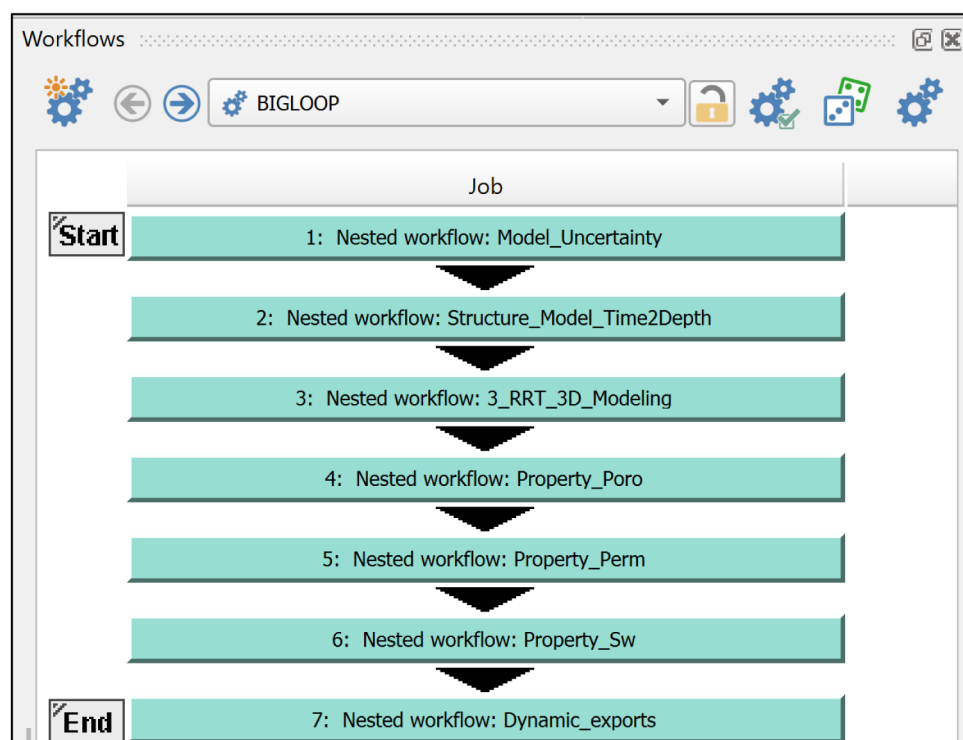


Figure 1—Repeatable geological modelling workflow

The entire workflow (Figure 2) is orchestrated using a commercially available platform for assisted history matching and uncertainty analysis. This platform enables the seamless generation of both geological and dynamic model realizations. The process begins with the submission of an initial batch of model realizations, the primary purpose of which is to explore and characterize the uncertainty solution space. The sampling of uncertainty variable values is carried out using an optimized version of the Latin Hypercube Sampling (LHS) technique, which ensures broad and efficient coverage of the defined uncertainty ranges. The results from this initial scoping phase are compared against historical production and pressure data, including well production rates and pressure observations, to validate the baseline model and the defined uncertainty configurations. Once the base validation is confirmed, a more targeted history matching phase begins.

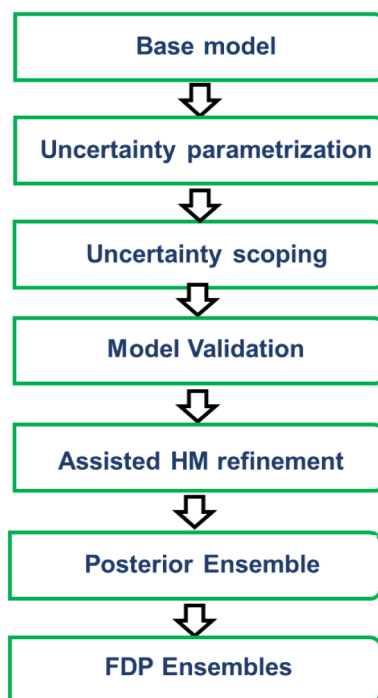


Figure 2—Integrated assisted HM and uncertainty analysis workflow

To refine the history match, objective functions are established by identifying key simulation variables of interest—such as static reservoir pressures, gas production rates, bottom-hole pressures, and Repeat Formation Test (RFT) pressures—along with their corresponding tolerances for each well. The results from the initial scoping runs are then used to construct sets of proxy models, or response surfaces, which approximate the behaviour of the reservoir system in a computationally efficient manner. These proxy models are integrated into the orchestration engine, which uses them to guide the selection of uncertainty variable values for subsequent refinement batches. The proxy models are continuously updated and retrained after each batch of simulations using a machine learning-based approach, thereby improving their predictive accuracy over time. This iterative process continues until multiple realizations achieve an acceptable history match, based on the predefined objective function criteria.

The likelihood of each reservoir model realization is then evaluated within a Bayesian framework. In this framework, prior distributions of the uncertainty variables are updated by assimilating observed historical data, resulting in posterior probability distributions that reflect the most likely parameter values (Fillacier S. et al.). These posterior distributions are sampled using a Markov Chain Monte Carlo (MCMC) algorithm to generate an ensemble of history-matched reservoir models. This ensemble forms the basis for subsequent forecasting, optimization, and decision-making under uncertainty.

Reservoir Uncertainties

As part of the modeling workflow, all available data across the various subsurface domains were thoroughly reviewed and subjected to rigorous quality control to identify key sources of uncertainty in the reservoir model. This process allowed for the definition of critical uncertainty parameters and their plausible ranges, forming the basis for a comprehensive uncertainty analysis.

Structural uncertainty

Structural model uncertainty was addressed using a full Horizon Uncertainty Modeling (HUM) from time to depth domain. In this workflow, the complete process of converting seismic interpretation from the time domain to the depth domain is addressed using the COHIBA integrated in static modelling software (Figure

3). The objective is to quantify and propagate uncertainties from seismic interpretation, velocity modeling, and structural complexity, ultimately delivering a set of geologically plausible depth models for further static and dynamic reservoir modeling. A reference depth model represents the basis for uncertainty quantification and propagation is initially constructed using:

- Six interpreted seismic time horizons representing key stratigraphic markers and structural boundaries within the study area
- Interpreted formation tops from wells
- Isochore maps derived from well data
- Average interval velocities

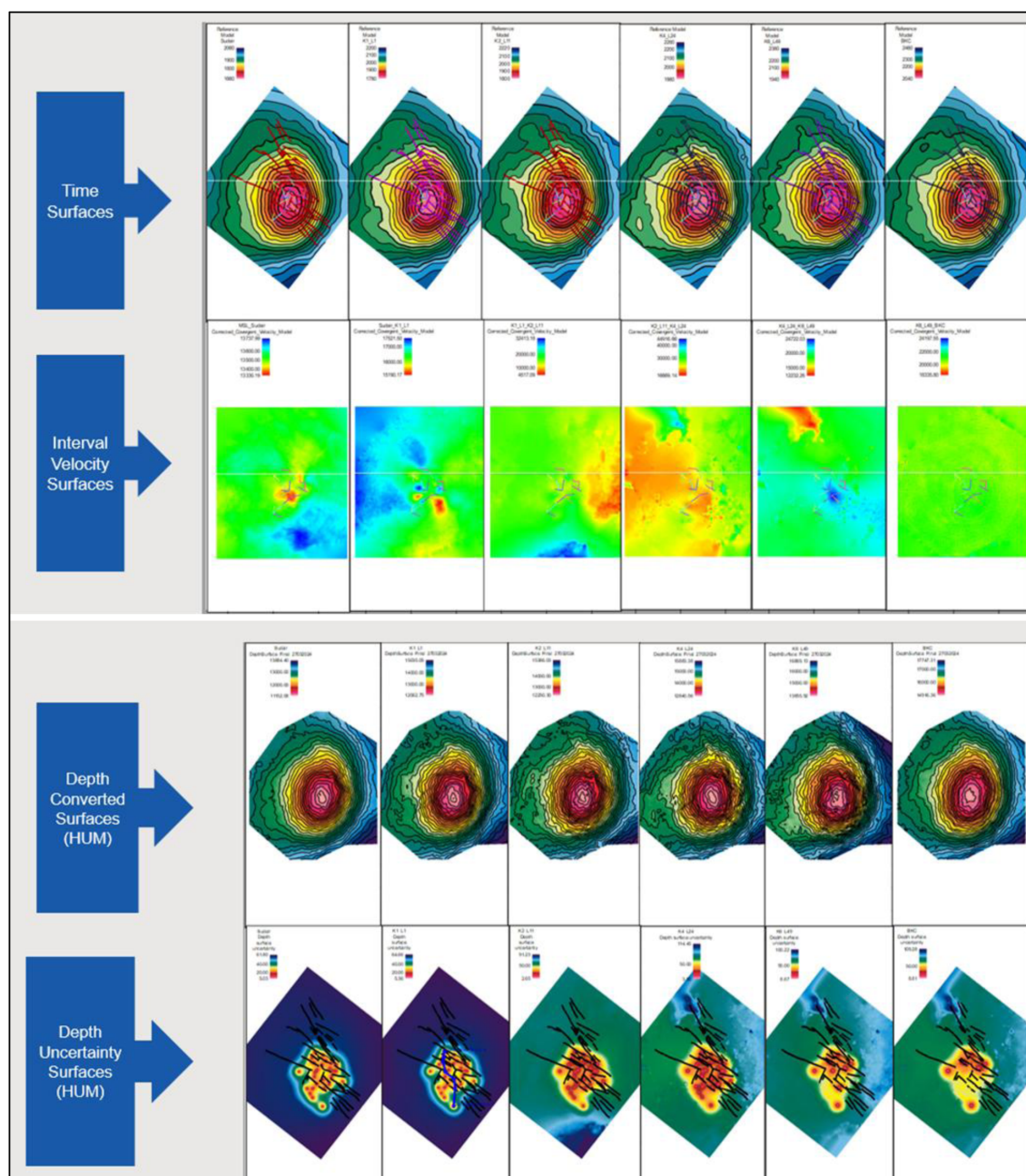


Figure 3—Structure uncertainty workflow implemented fully from time to depth domains using COHIBA

To generate a consistent set of depth-converted horizons that reflect geological realism and account for potential sources of uncertainty, the COHIBA workflow is employed. This involves introducing uncertainty envelopes to the following key inputs:

- Seismic horizon picks: To capture interpretation uncertainty, especially in zones of poor seismic resolution or structural complexity.
- Interval velocity models.
- Isochore trends: Particularly for thin stratigraphic units not visible on seismic

The workflow also allows for conditioning the depth-converted horizons to available well tops, ensuring the generated realizations remain geologically plausible and consistent with hard data. The outcome is a suite of multiple realizations of each structural horizon in depth. One uncertainty variable is used in the model to address the structure of these six different horizons together to avoid any unjustified thinning or thickening (Figure 4).

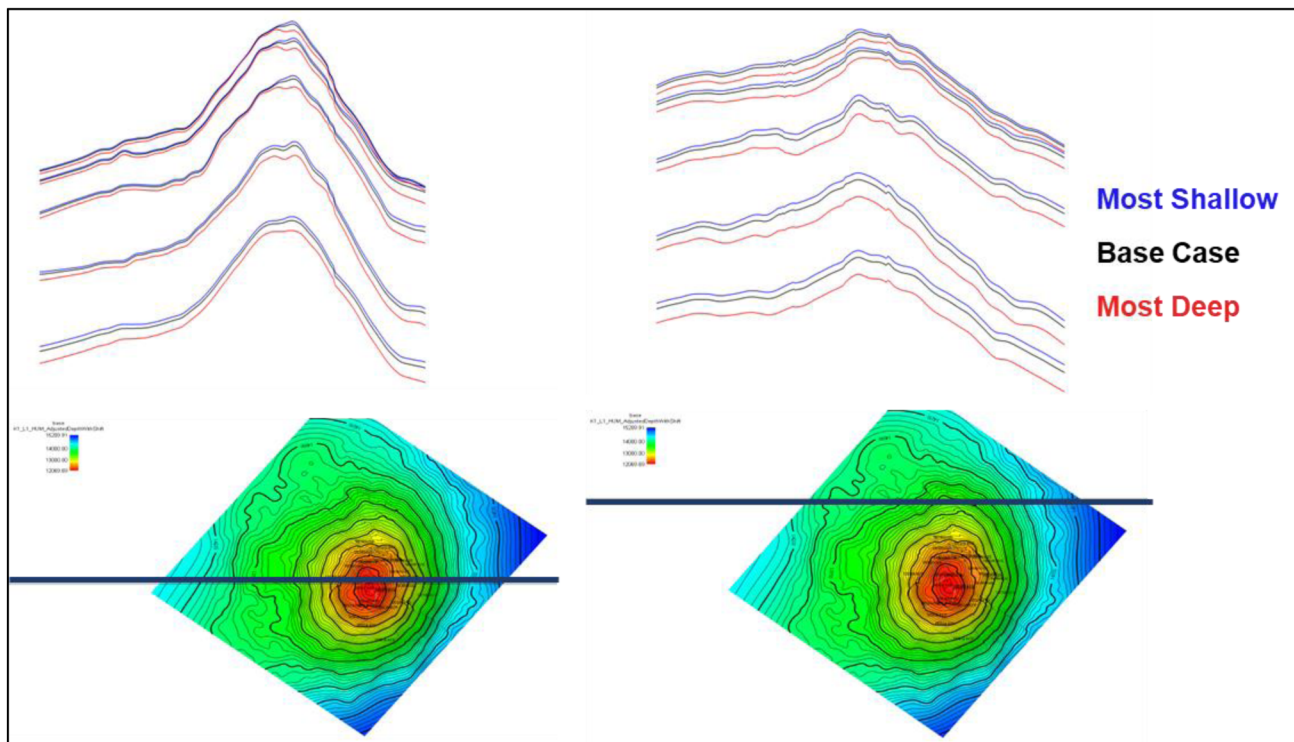


Figure 4—Two X-sections show the impact of structure uncertainty for the six different horizons in depth domain. blue is most shallow, black is base case and red is most deep case.

Facies Distribution Uncertainty

Facies distribution plays a fundamental role in subsurface reservoir modeling, as it governs the spatial distribution of geological rock types and, consequently, the heterogeneity of the reservoir. All subsequent petrophysical properties—such as porosity and permeability—are typically distributed within the 3D cellular grid bias to the facies model, making the facies model a primary driver of the reservoir properties in the model.

Based on geological investigations, including core analysis, petrophysical interpretation and sedimentological interpretation, nine distinct rock types have been identified in the field (Figure 5). These are grouped as follows:

- Dolomite: Five different rock types, ranging from D1 (representing the highest quality dolomite with superior reservoir characteristics) to D5 (the poorest quality).
- Limestone: Three rock types, from L1 (highest quality) to L3 (lowest quality).
- Anhydrite: A dense, inactive reservoir rock type

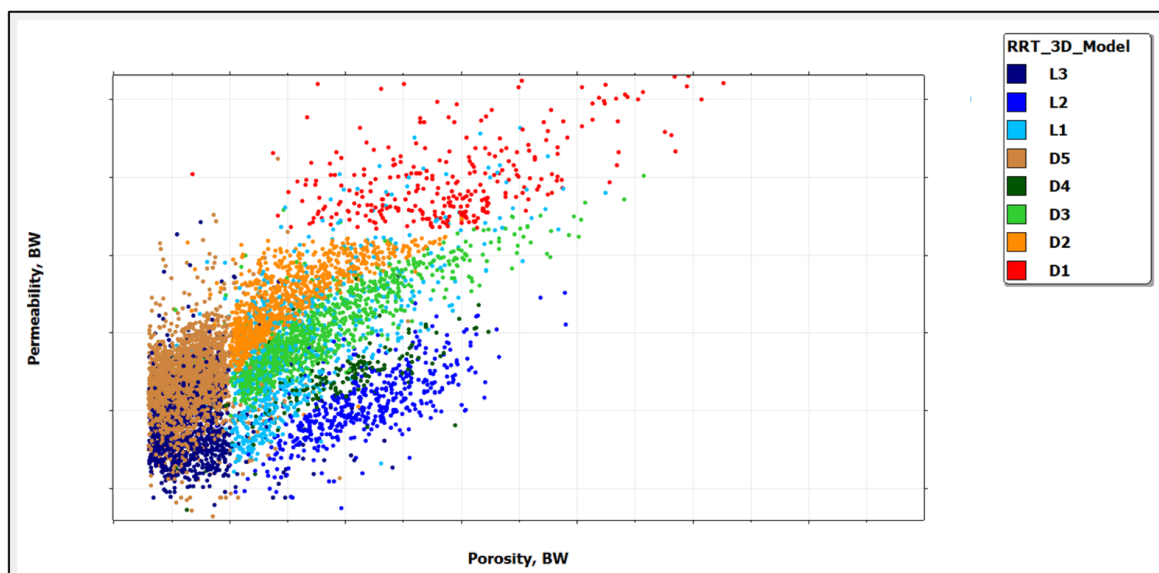


Figure 5—Porosity permeability cross blot for well-blocked data, different colours for different rock types

Well log interpretation and facies analysis in the drilled wells reveal a consistent trend of decreasing rock quality with increasing depth. This degradation is particularly evident in the flank areas of the field, where low-quality facies types D5 and L3 are dominant. In the absence of a detailed seismic inversion study capable of delineating facies trends laterally and vertically, this degradation trend was addressed by defining spatial polygons for each of the five reservoir layers (Figure 6). These polygons represent zones of better rock quality, while the areas outside are assumed to be dominated by the lower quality D5 and L3 facies in addition to Anhydrite. Consequently, five uncertainty parameters were introduced—one for each reservoir—to capture the uncertainty in the spatial extent of facies degradation.

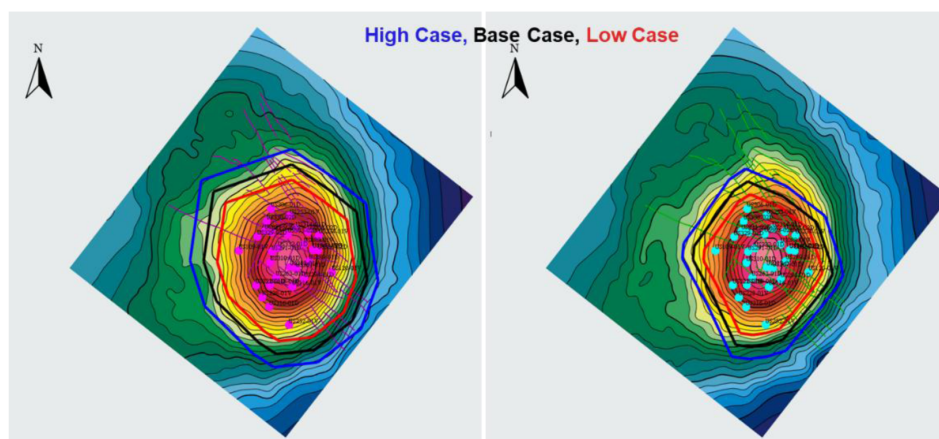


Figure 6—Structure maps for top two reservoirs showing the facies quality polygon uncertainties. Blue is high case; Red is low case and black is the base case.

In addition to the vertical degradation trend, the spatial continuity and distribution of facies within each reservoir were treated as a separate source of uncertainty. This was quantified using ten additional uncertainty parameters, allowing variation in the variogram ranges (major, minor, and vertical), two variables for each reservoir. These parameters influence how facies distributed within the model and, in turn, affect the resulting heterogeneity of the **Petrophysical Porosity Uncertainties**

Porosity is distributed within the model using stochastic algorithms bias to the previously defined Reservoir Rock Types (RRTs). Petrophysical porosity logs from wells are first upscaled to the model grid scale through a well-blocking process. The statistical characteristics of these blocked logs - such as mean, standard deviation, skewness, etc.- are then extracted and used to inform the stochastic porosity distribution. To account for uncertainty in the petrophysical porosity logs, five uncertainty parameters were defined. These parameters varies the blocked porosity logs for different reservoir resulting in different porosity distributions in different model realization (Figure 7).

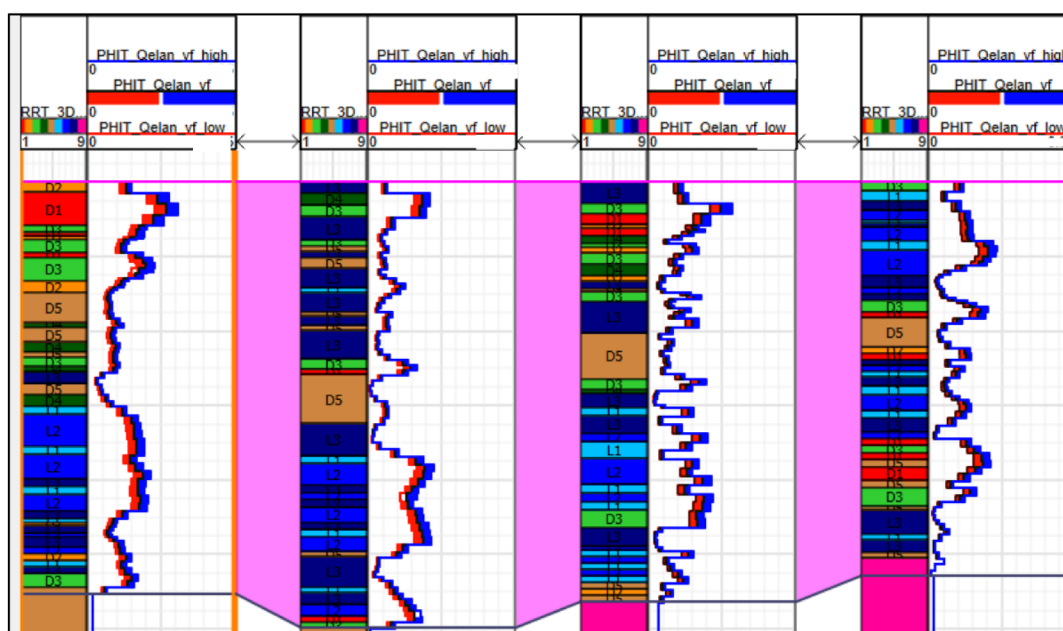


Figure 7—Petrophysical blocked-log uncertainty for four wells, top reservoir as an example. blue represents the high side uncertainty, red represents the low side and black is the base case

To ensure consistency with measured data, later in the workflow the porosity values at existing well locations are preserved using a local update mechanism within the static modeling platform. This ensures that the modeled porosity honors well log observations precisely at well locations, which is particularly important for initial water saturation distribution (Initial water saturation is distributed via J-Functions, those J-Functions are function of porosity). Maintaining accurate porosity at well control points is therefore critical to the reliability of the subsequent dynamic model.

Petrophysical Permeability Uncertainties

Permeability distribution within the dynamic model is typically bias to facies and statistically correlated with porosity. Similar to porosity modeling, permeability input data is derived from petrophysical logs, which are processed and upscaled (blocked) at well locations to be integrated into the model.

To enhance the understanding of spatial permeability variations, a detailed analysis of dynamic well performance (covering pressure data, production history, and Production Logging Tool (PLT) surveys) was conducted for wells completed in the top four reservoir units. This analysis revealed distinct differences in dynamic behavior that allowed the classification of wells into seven areally-defined groups,

each demonstrating unique reservoir performance characteristics. These performance differences reflect heterogeneities in reservoir permeability across the model area. In order to represent these observed areal and vertical variations in permeability, a total of 28 uncertainty variables were introduced for the top four reservoirs (7 areal groups $\times$ 4 reservoirs). Furthermore, an additional permeability uncertainty variable was defined for the fifth, deeper isolated reservoir unit, which is treated independently due to its unique well performances (Figure 8).

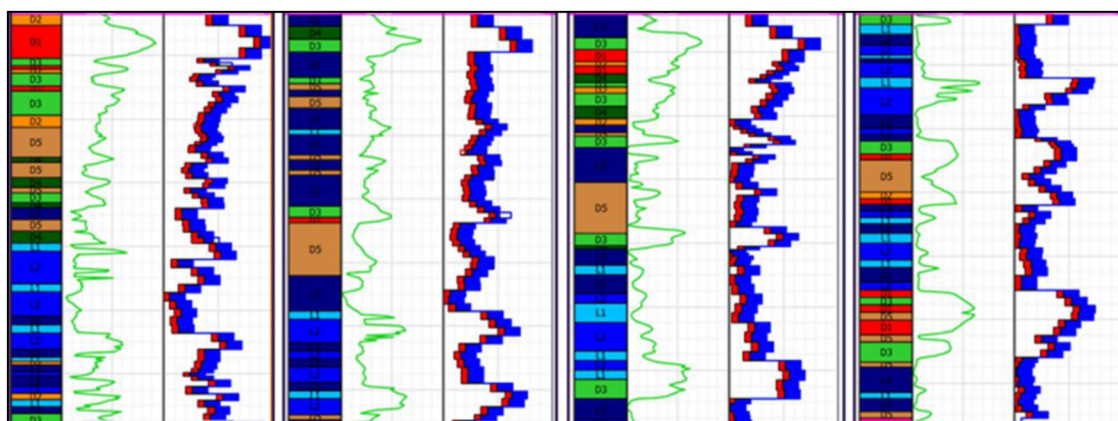


Figure 8—Petrophysical Permeability uncertainty for four wells and top reservoir as an example. First track blocked facies logs, second track with green curve is petrophysical log, and third track shows the blocked permeability uncertainty (red shows the low side, blue shows the high side, and black is the base curve)

Additional Model Uncertainties

Further sources of geological and dynamic uncertainty were identified through comprehensive review and integration of all available subsurface data.

- Gas-Water Contact (GWC) Uncertainty: The fifth and deepest reservoir unit exhibits uncertain gas-water contact depth. This uncertainty is captured through a dedicated variable to assess its potential impact on volumetrics and well performance in that interval.
- Inter-Reservoir Communication: The upper four reservoir units are known to be hydrodynamically connected, yet the degree of vertical communication between them remains uncertain. To address this, three transmissibility multiplier uncertainty variables are implemented to model the range of possible pressure communication scenarios between these stacked reservoirs.
- Vertical-to-Horizontal Permeability Ratio (K_v/K_h): Accurate representation of anisotropy is crucial for dynamic flow simulation. To incorporate the uncertainty in vertical permeability behavior, five separate K_v/K_h uncertainty variables are defined (one for each of the five reservoir zones) to capture the range of plausible vertical transmissibility conditions.
- Relative permeability uncertainty: one uncertainty variable was used to vary the gas relative permeability end point for each SATNUM region (Figure 9)

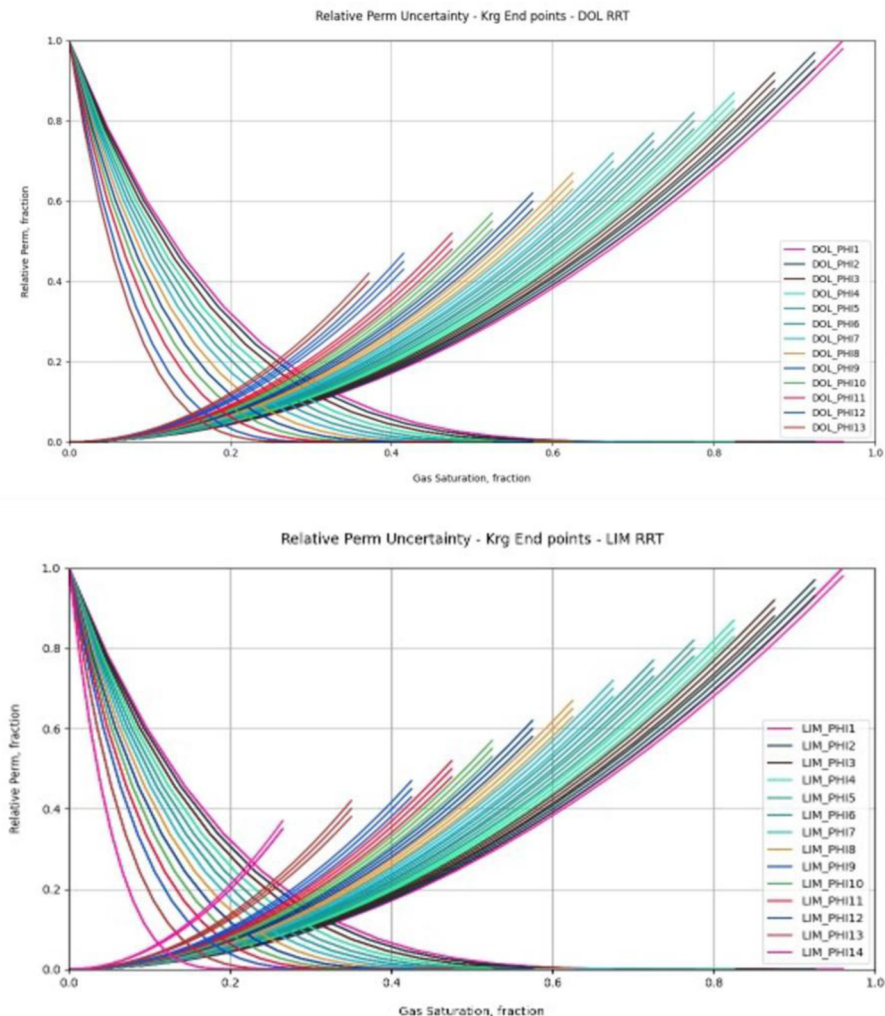


Figure 9—shows the uncertainty in Krg end point for each SATNUM region in the model. Oil-Gas relative permeability for Dolomite group (13 SATNUM regions) in the upper plot and Limestone group (14 SATNUM regions) in the lower plot.

Results

The integrated workflow outlined in the methodology section has been successfully applied to evaluate the overall field history matching and calibration process. As a result, a posterior ensemble comprising 100 simulation cases was generated, each representing a unique combination of the defined uncertainty variables. These cases collectively form a comprehensive representation of the possible geological and reservoir behavior under the defined uncertainty space. Importantly, every case within the posterior ensemble maintains full geological consistency, meaning that no manual or non-justified adjustments were introduced during the process. Each model is backed by a complete, traceable geological realization that aligns with the initial structural and property modeling, ensuring high degree of model reliability and reproducibility of the results.

The performance of the history matching process was evaluated using a variety of quantitative quality indicators, both at the individual well level and across the entire field. Key metrics include maximum match quality, average match quality, and the half sum of squared differences between observed and simulated production or pressure data ensuring a robust evaluation of model performance. Figure 10 below illustrates the evolution of the average overall quality, which is defined as the absolute difference between simulated and observed data, weighted by their respective allowable tolerance ranges. Lower values on this metric indicate a better history match.

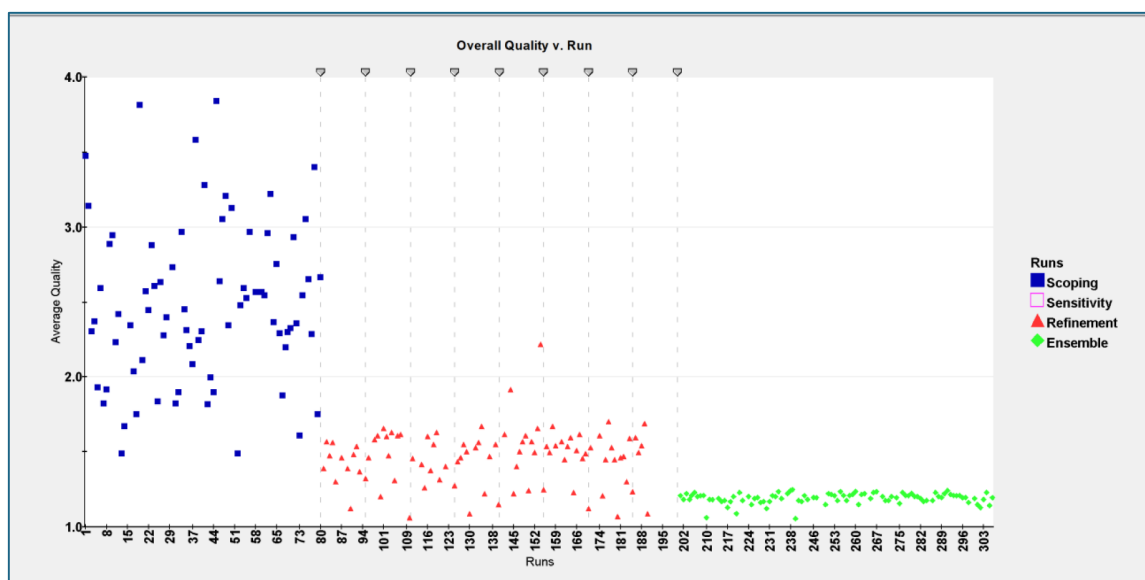


Figure 10—Average overall quality evolution plot. Blue dots represent scoping cases sampled using optimized LHC, Red dots represent 8 different batches of history match refinement where proxy models and machine learning are used to improve history match qualities, green dots represent the final posterior ensemble using Bayesian updates.

The following figures present selected results from the final posterior ensemble history matching process, focusing on a subset of wells and key production performance indicators. **Figure 11** displays the history match of static reservoir pressure across several representative wells, highlighting the quality of history match for all ensemble cases and variability captured by the ensemble approach. **Figure 12** presents the matched results for gas production rates for the same wells, while **Figure 13** illustrates the match quality for tubing head pressure, offering additional insight into the dynamic behavior of the reservoir system under the defined uncertainty conditions.

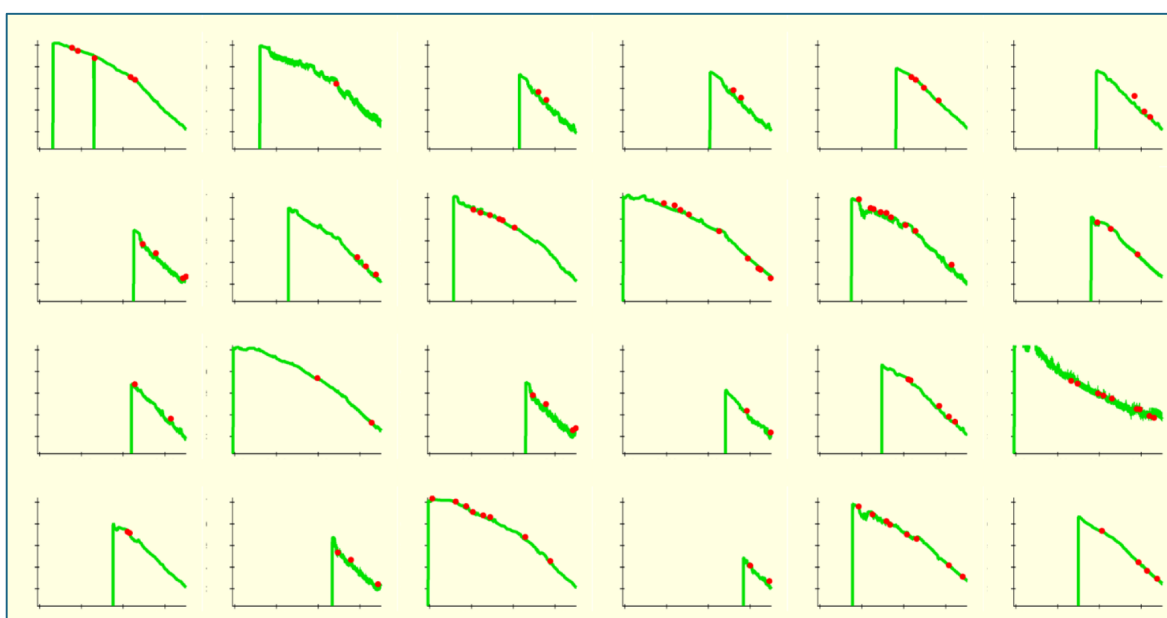


Figure 11—shows the Static pressure match for few wells in the model versus time, red dots are observed data and green lines are final posterior ensemble cases

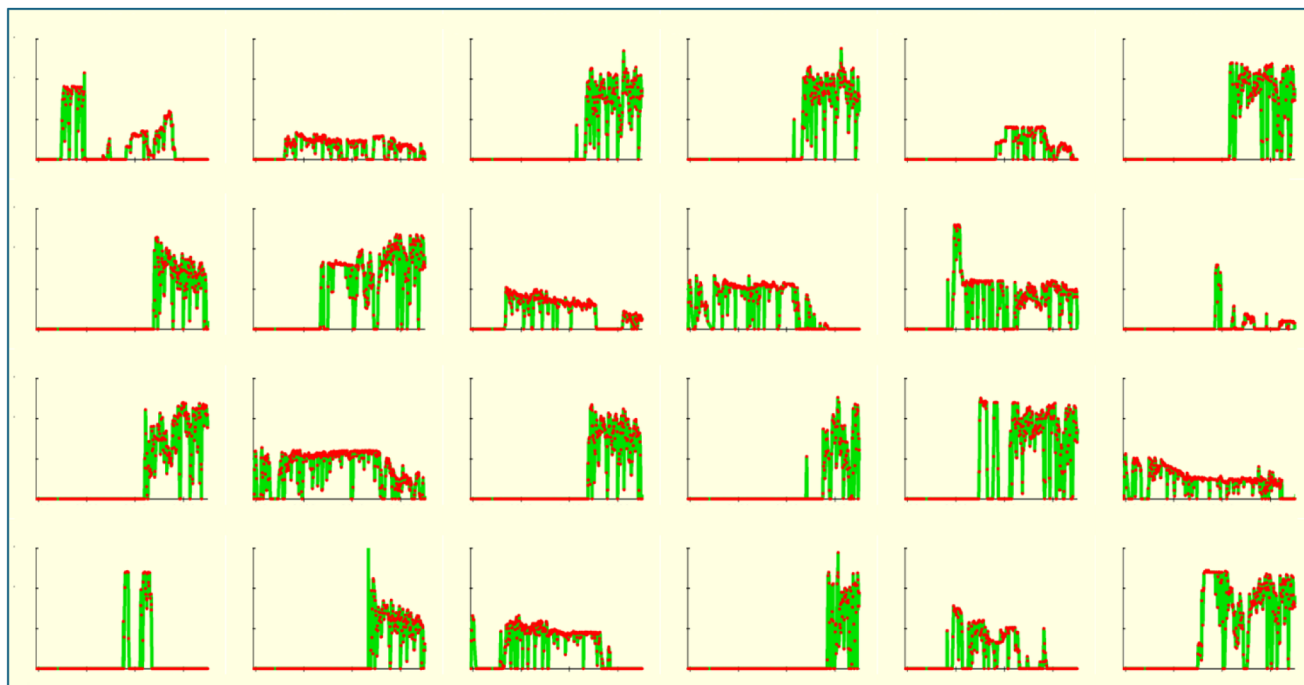


Figure 12—shows the Gas production rate match for few wells in the model versus time, red dots are observed data and green lines are final posterior ensemble cases

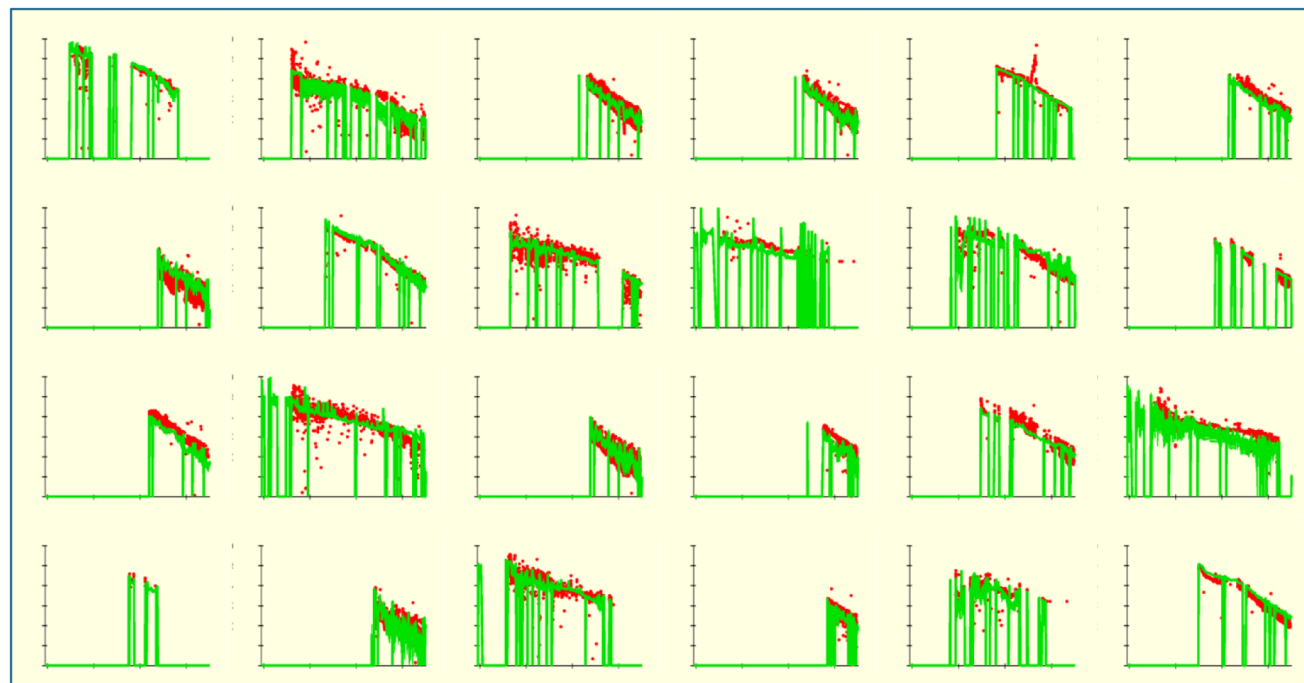


Figure 13—shows the Tubing head pressure match for few wells in the model versus time, red dots are observed data and green lines are final posterior ensemble cases

Prediction Phase and Uncertainty Quantification

After successful history matching, the final posterior ensemble was carried forward into the forecasting (prediction) phase to quantify the range of potential outcomes and associated uncertainties. This propagation enables a rigorous assessment of key forecast parameters, including expected gas recovery factors, plateau duration, and cumulative gas production.

The complete posterior ensemble consisting of 100 geologically consistent realizations was used to simulate and compare the outcomes of three distinct Field Development Plans (FDP). For confidentiality reasons, results and plots of these prediction scenarios are not published in this paper:

- No Further Action FDP
- The current Business Plan FDP
- Field Potential FDP

By leveraging the ensemble approach instead of relying on a single deterministic model, the analysis captures a wide range of possible reservoir behaviors, driven by subsurface uncertainties. This allows for a probabilistic evaluation of gas recovery potential over time, providing valuable insight into the likelihood of meeting production targets and enabling more informed, risk-aware decision-making regarding development strategies.

The ensemble-based forecast supports robust risk mitigation, scenario comparison, and value-of-information assessment, which are essential for optimizing field development under geological uncertainty.

Conclusions

The workflow presented in this paper has been successfully applied to the history matching of a mature offshore gas field. The implementation has demonstrated a significant improvement in history match quality, achieved without compromising geological consistency.

This study highlights the critical importance of integrating multiple subsurface modeling domains (structural, facies, petrophysical, and dynamic) and ensuring that uncertainties are propagated coherently across all domains. The methodology ensures that every model realization remains geologically consistent, reinforcing the credibility of the model.

Utilizing the full posterior ensemble during the prediction phase provides a comprehensive view of potential reservoir performance. This approach enables the estimation of production probabilities across various timeframes and allows for a realistic assessment of different Field Development Plans. It enhances the ability to quantify and mitigate risk, supporting more informed and resilient development strategies under geological uncertainty.

Furthermore, the automation of the workflow and the incorporation of machine learning-based proxy models within the history matching process have significantly reduced the study's execution time. These efficiencies not only accelerate model iteration cycles but also facilitate broader exploration of the uncertainty space.

The seamless integration between various subsurface domains has resulted in an evergreen modeling framework, where new data can be easily incorporated. This enables continuous model updates with minimal effort, ensuring the reservoir model remains current and decision-ready throughout the field's lifecycle.

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References

- Alqallabi S., Abd-Allah M., Bahussain E., Al-Jenaibi F., Parra H., Shebl H., Altaf B., Nehaid H., Fathy A., Al Menterhi K., (2020) "An Integrated Approach to Quantify Reservoir Uncertainties for a Reliable Production Forecast and FasterModelling Process for Offshore Field, Abu Dhabi, UAE" SPE-203429-MS
- Abdallah M, Abdelrahman A, Brul L, Taha T, Javed M. "Probabilistic Economical Evaluation for Business Decisions ThroughIntegrated Uncertainty Assessment and Reliable Ensemble-BasedProduction Forecasts" SPE-207317-MS

Fillacier S., Fincham A. E., Hammersley R. P., Heritage J. R., Kolbikova I., Peacock G. and Soloviev V.Y., (2014).
"Calculating Prediction Uncertainty using Posterior Ensembles Generated from Proxy Models" SPE-171237-MS.